

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

JANUARY EXAMINATION

BACS2163 SOFTWARE ENGINEERING

WEDNESDAY, 8 JANUARY 2025

TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN INTERACTIVE SOFTWARE
TECHNOLOGY

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

BACHELOR IN DATA SCIENCE (HONOURS)

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BACS2163 SOFTWARE ENGINEERING**Question 1**

- a) (i) Fred Brooks' assertion "*plan to throw one away – you will anyway*" highlights flaws in the waterfall model of software development. Identify **TWO (2)** specific disadvantages of the waterfall model that he is referring to. (4 marks)
- (ii) Propose an alternative software process model for Question 1 a) (i) and explain **TWO (2)** ways this model addresses the identified disadvantages. (1 + 4 marks)
- b) Assume you are developing a mobile banking application. Identify **THREE (3)** examples of non-functional requirements for each the following categories based on this scenario:
- (i) Product Requirements (3 marks)
- (ii) Organisational Requirements (3 marks)
- (iii) External Requirements (3 marks)
- c) Which maintenance type would be applied to fix the issue in the E-Math application that incorrectly calculates the area of a circle when the radius is provided. Provide **THREE (3)** reasons for your answer. (1 + 6 marks)

[Total: 25 marks]

Question 2

- a) A bank plans to develop a new mobile banking application to enhance customer service. For each of the following functional requirements, identify and describe any **THREE (3)** non-functional requirements.
- (i) Account balance checks (6 marks)
- (ii) Fund transfers (6 marks)
- (iii) Transaction history (6 marks)
- b) An online payment processing system for an e-commerce platform is designed to handle user transactions in real-time. The system must process payment requests, validate credit card details, and complete the transaction within a specific time frame to prevent delays. If the system fails to process a transaction within the required time, it can result in an inconvenience for the user, but the transaction can be retried or postponed without serious consequences. Analyse any **ONE (1)** suitable category of real time system and provide **THREE (3)** justifications explaining why it is the most appropriate for this scenario. (7 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 3**

- a) A large e-commerce application is being developed, and several modules (e.g., Product Catalog, Payment Gateway, and Order Management) have been completed. The development team is now focused on integrating these modules to ensure seamless functionality. Analyse **ONE (1)** suitable integration testing technique and provide **THREE (3)** explanations to support your choice. (7 marks)
- b) In software architecture, the Repository Model, Layered Model, and Client-Server Model are widely used for system design and structuring. Compare these models based on their approaches to data management and system interaction. Provide **THREE (3)** key points for each model under both criteria. (18 marks)

[Total: 25 marks]

Question 4

- a) How do the User Interface (UI) design principles of consistency, user guidance, and recoverability apply to design a dashboard for an online banking application? Provide **THREE (3)** specific examples for each principle to support your response. (9 marks)
- b) In the context of a mid-sized software company developing a new mobile banking application within six months, identify and explain the **FOUR (4)** key responsibilities that a project manager should focus on to ensure the project's success. (16 marks)

[Total: 25 marks]